

SOBOLEV REMOVABILITY OF HÖLDER GRAPHS AT THE CRITICAL EXPONENT

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ABSTRACT. We prove that every α -Hölder graph is removable for continuous $W^{1,p}$ functions at the critical exponent $\alpha = p/(2p - 1)$, for $1 \leq p < \infty$, and at $\alpha = 1/2$ for $p = \infty$. In particular, every $2/3$ -Hölder graph is conformally removable.

AI USE STATEMENT

This note, including its proofs, was generated almost entirely by GPT-6 Pro, with only minor revisions to the exposition made by the author. The author has carefully checked all proofs, confirms their correctness, and takes full responsibility for the contents of the note.

1. INTRODUCTION AND RESULTS

A compact set $E \subset \mathbb{R}^2$ is *removable for continuous $W^{1,p}$ functions* if, for every open neighborhood D of E ,

$$u \in C(D) \cap W^{1,p}(D \setminus E) \implies u \in W^{1,p}(D).$$

We use this convention throughout, as in Jones–Smirnov [JS00] and Tecu [Tec10].

Kaufman constructed non-removable graphs in [Kau84] and studied their dependence on the modulus of continuity in [Kau86]. His $r = 1$ removability theorem assumes that first derivatives off the graph admit continuous extensions across it [Kau86, Theorem 1]. For $1 < p < \infty$, Tecu [Tec10, Theorems 3–4] gave an integral sufficient condition on the modulus of continuity, which yields removability of α -Hölder graphs when $\alpha > p/(2p - 1)$ via the Jones–Smirnov criterion, and constructed non-removable examples when $\alpha < p/(2p - 1)$. Tecu also records a theorem of Jones [Tec10, Theorem 6] giving removability of α -Hölder graphs for continuous $W^{1,\infty}$ functions when $\alpha > 1/2$. The construction in [Tec10, proof of Theorem 5] produces a continuous function whose weak gradient belongs to L^∞ off the graph but which fails absolute continuity on lines, giving non-removable examples for continuous $W^{1,\infty}$ functions when $\alpha < 1/2$.

Theorem 1.1. *Let $1 \leq p \leq \infty$, and define*

$$\alpha_p = \frac{p}{2p - 1} \quad (1 \leq p < \infty), \quad \alpha_\infty = \frac{1}{2}.$$

If $f : [0, 1] \rightarrow \mathbb{R}$ is α_p -Hölder continuous, then its graph $\Gamma_f = \{(x, f(x)) : 0 \leq x \leq 1\}$ is removable for continuous $W^{1,p}$ functions.

For $1 < p < \infty$, this settles the equality case left open in [Tec10]. For $p = \infty$, it establishes removability for every $1/2$ -Hölder graph. The case $p = 1$ concerns Lipschitz graphs and is included in the same proof.

Together with Tecu’s examples, the theorem gives the sharp Hölder threshold for continuous Sobolev removability. For $p = 1$, sharpness below $\alpha_1 = 1$ follows by choosing $q > 1$ sufficiently

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close to 1 that $\alpha < q/(2q - 1)$ and using the implication from $W^{1,1}$ -removability to $W^{1,q}$ -removability. Indeed, on a bounded neighborhood of an area-zero set, a continuous $W^{1,q}$ function off the set that extends as a $W^{1,1}$ function has the same weak derivatives almost everywhere; these derivatives remain in L^q .

The standard implication from continuous $W^{1,2}$ -removability to conformal removability [JS00] gives the following consequence.

Corollary 1.2. *Every $2/3$ -Hölder graph is conformally removable.*

For comparison, Doherty–Miller [DM26] proved that the graph of one-dimensional Brownian motion is almost surely removable for continuous $W^{1,\infty}$ functions but non-removable for continuous $W^{1,p}$ functions for every finite p . They also proved $W^{1,\infty}$ -removability of $1/2$ -Hölder graphs under the additional assumption that the $3/2$ -Hausdorff measure of Γ is zero (see the paragraph before Lemma 3.4 therein); Theorem 1.1 removes this assumption. Brownian paths are almost surely not $1/2$ -Hölder, so their $W^{1,\infty}$ -removability theorem is not subsumed by Theorem 1.1.

The proof for finite p uses absolute continuity of the integral at $\alpha = p/(2p - 1)$. The proof for $p = \infty$ instead combines a parabolic blow-up with a rigidity lemma for occupation densities. Both arguments are local and apply equally to locally Hölder graphs over arbitrary intervals.

2. DETOURS AND THE CASE FOR FINITE p

We first describe a localization that will be used throughout. Extend f constantly past 0 and 1; this preserves its Hölder constant. Given a function in the definition of removability, multiply it by a smooth cutoff function compactly supported in its neighborhood and equal to one near Γ_f , then extend by zero. It suffices to prove the result for

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad u \in C_c(\mathbb{R}^2), \quad u \in W^{1,p}(\mathbb{R}^2 \setminus \Gamma), \quad \Gamma = \{(x, f(x)) : x \in \mathbb{R}\}.$$

Once the cutoff function extends, the original function has its original, L^p -integrable derivatives across Γ_f as well. The enlarged graph causes no difficulty: away from the original compact graph the cutoff function already belongs to $W^{1,p}$. Since Γ has area zero, let $a, b \in L^p(\mathbb{R}^2)$ denote the weak derivatives u_x, u_y off Γ , extended by zero on Γ .

Almost every vertical restriction of u is absolutely continuous on the two sides of its single intersection with Γ . On bounded intervals, its derivative $b(x, \cdot)$ is integrable. Taking the limit at the intersection and using continuity of u glues the two restrictions into an absolutely continuous function. The slicing characterization therefore gives

$$D_y u = b \, dx \, dy. \tag{2.1}$$

For $p = \infty$, if $|b| \leq M$, these restrictions are M -Lipschitz. Continuity of u then gives the same bound on *every* vertical line by approximation in the horizontal variable.

Fix $K \geq 1$ such that

$$|f(x) - f(x')| \leq K|x - x'|^\alpha.$$

Choose bounded intervals $I \Subset I^+$ with the horizontal projection of $\text{supp } u$ compactly contained in I . For small $h > 0$, set $H = Kh^\alpha$ and define

$$B_h = \{(x, y) : x \in I, |y - f(x)| \leq H\}, \quad N_h = \{(x, y) : x \in I^+, |y - f(x)| \leq 5H\}.$$

Write

$$q_h u(x, y) = \frac{u(x + h, y) - u(x, y)}{h}, \quad A_h a(x, y) = \frac{1}{h} \int_x^{x+h} a(t, y) \, dt.$$

For almost every $(x, y) \notin B_h$ with $x \in I$, the horizontal segment between (x, y) and $(x + h, y)$ is disjoint from Γ , and hence

$$q_h u(x, y) = A_h a(x, y). \tag{2.2}$$

Lemma 2.1. *With this notation,*

$$\int_{I \times \mathbb{R}} |q_h u - A_h a| dx dy \leq \frac{4H}{h} \int_{N_h} |b| dx dy + 3 \int_{N_h} |a| dx dy.$$

Proof. For almost every $(x, y) \in B_h$, join (x, y) to $(x + h, y)$ by two vertical segments and a horizontal segment at height

$$s \in [f(x) + 2H, f(x) + 3H].$$

The horizontal segment lies above Γ . Set

$$\beta_h(x) = \int_{f(x)-5H}^{f(x)+5H} |b(x, z)| dz.$$

Averaging over the indicated heights s and using (2.1) for the vertical segments gives

$$h|q_h u(x, y)| \leq \beta_h(x) + \beta_h(x + h) + \frac{1}{H} \int_{f(x)+2H}^{f(x)+3H} \int_x^{x+h} |a(t, s)| dt ds.$$

Integration in $(x, y) \in B_h$ and Fubini's theorem now give

$$\int_{B_h} |q_h u| dx dy \leq \frac{4H}{h} \int_{N_h} |b| dx dy + 2 \int_{N_h} |a| dx dy. \quad (2.3)$$

Here each vertical segment is contained in N_h , and integration over the starting heights y contributes a factor of $2H$. For the horizontal segment, averaging contributes H^{-1} , integration in y contributes $2H$, and for each intermediate horizontal coordinate t the possible starting points x form an interval of length at most h . For these intermediate points, $|s - f(t)| \leq 4H$, as required. Division by h gives the second term.

The same Fubini argument for horizontal segments gives

$$\int_{B_h} |A_h a| dx dy \leq \int_{N_h} |a| dx dy.$$

Combining this with (2.2) and (2.3) proves the lemma. $\square$

Proof of Theorem 1.1 for $p < \infty$. Since $\text{Area}(N_h) \leq CH$, Hölder's inequality gives

$$\frac{H}{h} \int_{N_h} |b| dx dy \leq C \frac{H^{2-1/p}}{h} \|b\|_{L^p(N_h)}.$$

For $\alpha = p/(2p - 1)$ one has $\alpha(2 - 1/p) = 1$. Since $H = Kh^\alpha$, the coefficient on the right is bounded independently of h . Since $p < \infty$ and $\text{Area}(N_h) \rightarrow 0$,

$$\|b\|_{L^p(N_h)} \rightarrow 0, \quad \int_{N_h} |a| dx dy \rightarrow 0.$$

Also $A_h a \rightarrow a$ in $L^1(I \times \mathbb{R})$. Thus Lemma 2.1 yields $q_h u \rightarrow a$ in L^1 . The functions $q_h u$ are supported in $I \times \mathbb{R}$ for small h , and distributional convergence identifies $D_x u = a dx dy$. Together with (2.1), this proves $u \in W^{1,p}(\mathbb{R}^2)$. The calculation includes $p = 1$. $\square$

3. A RIGIDITY LEMMA FOR OCCUPATION DENSITIES

For a continuous path $g : J \rightarrow \mathbb{R}$, its occupation measure on $[s, t] \subset J$ is the measure

$$A \mapsto \int_s^t \mathbf{1}_A(g(r)) dr.$$

In the next lemma, a continuous density means a density defined and continuous on all of $\mathbb{R}$, including the endpoints of its support. For a function $v : \mathbb{R} \rightarrow \mathbb{R}$, write $\text{Lip}(v) := \sup_{x \neq y} \frac{|v(x) - v(y)|}{|x - y|} \in [0, +\infty]$.

Lemma 3.1. *Let J be a nondegenerate open interval. There are no continuous functions $g : J \rightarrow \mathbb{R}$ and $V : J \times \mathbb{R} \rightarrow \mathbb{R}$ for which all of the following hold with finite constants $K > 0$ and $M \geq 0$:*

$$\begin{aligned} |g(t) - g(s)| &\leq K\sqrt{|t - s|}, \\ \text{Lip}(V(t, \cdot)) &\leq M \quad (t \in J), \\ V(t, \cdot) - V(s, \cdot) &\text{ is a continuous occupation density of } g \text{ on } [s, t] \quad (s < t). \end{aligned} \quad (3.1)$$

Proof. Suppose that such g and V exist. We write $L_{s,t}(y) = V(t, y) - V(s, y)$. Every $L_{s,t}$ is continuous, nonnegative, and vanishes outside $g([s, t])$. We divide the proof into three steps.

Step 1: oscillation on both sides of almost every time. Fix $[q, T] \subset J$ and consider the set of running-minimum times

$$R_{q,T} = \{t \in [q, T] : g(t) = \min_{q \leq r \leq t} g(r)\}.$$

Its occupation measure is dominated by the occupation measure on $[q, T]$ and therefore has an integrable density $\rho \geq 0$. This density is supported in $[a, b]$, where $a = \min_{t \in [q, T]} g(t)$ and $b = g(q)$.

For $a < y < b$, let τ_y be the first hitting time of y after q . A running-minimum time whose value exceeds y must precede τ_y . Hence, as densities on (y, ∞) ,

$$\rho(z) \leq L_{q, \tau_y}(z) \quad \text{for almost every } z > y.$$

The path has no values below y on $[q, \tau_y]$, so continuity implies $L_{q, \tau_y}(y) = 0$. Consequently

$$\lim_{\varepsilon \downarrow 0} \frac{1}{\varepsilon} \int_y^{y+\varepsilon} \rho(z) dz = 0.$$

Lebesgue differentiation gives $\rho = 0$ almost everywhere on (a, b) . The endpoints $\{a, b\}$ contribute no atoms because the occupation measure has a density; if $a = b$, absolute continuity gives the same conclusion immediately. Thus $R_{q,T}$ has Lebesgue measure zero.

Taking the countable union over rational q, T proves that the set of left-sided local-minimum times has Lebesgue measure zero. Reflection of g in space, and reversal of time, give the other three one-sided local extrema. We may therefore choose an interior time t_* such that, in every sufficiently small neighborhood of t_* , there are values strictly above and strictly below $g(t_*)$ on *each* side of t_* . This step uses only the existence of the continuous densities.

Step 2: a crossing improves a spatial derivative bound. Suppose that, on an open rectangle $I \times Q \subset J \times \mathbb{R}$,

$$\text{Lip}_Q(V(t, \cdot)) \leq A \quad (t \in I), \quad A \geq 0. \quad (3.2)$$

Here $\text{Lip}_Q(v) := \sup_{x, y \in Q, x \neq y} \frac{|v(x) - v(y)|}{|x - y|}$. Fix $t \in I$ and a differentiability point $y \in Q$ of $V(t, \cdot)$. Assume $g(t) > y$ and that a later time $t_1 \in I$ satisfies $g(t_1) < y$. For all sufficiently small $r > 0$, the band $[y - r, y + r]$ lies in Q , $g(t) > y + r$, and $g(t_1) < y - r$. Let τ_r be the first time after t when $g = y - r$, and let σ_r be the last time in $[t, \tau_r]$ when $g = y + r$. Then

$$g([\sigma_r, \tau_r]) = [y - r, y + r], \quad \tau_r - \sigma_r \geq \frac{4r^2}{K^2}.$$

The nonnegative density $w_r = L_{t, \tau_r}$ satisfies $w_r(y - r) = 0$ by the first-hitting property and continuity. Moreover,

$$\int_{y-r}^{y+r} w_r(z) dz \geq \tau_r - \sigma_r \geq \frac{4r^2}{K^2}. \quad (3.3)$$

Put $d = V_y(t, y)$. The bound (3.2) at time τ_r and the equality $w_r(y - r) = 0$ give

$$w_r(z) \leq A(z - y + r) - (V(t, z) - V(t, y - r)), \quad y - r \leq z \leq y + r.$$

Differentiability at y implies, uniformly in this band,

$$V(t, z) - V(t, y - r) = d(z - y + r) + o(r).$$

Therefore

$$\int_{y-r}^{y+r} w_r(z) dz \leq 2(A - d)r^2 + o(r^2). \quad (3.4)$$

Comparing (3.3) and (3.4), we obtain

$$d \leq A - c, \quad \text{where } c = \frac{2}{K^2} > 0.$$

For an earlier time $t_0 < t$ such that $g(t_0) < y$, apply the same argument to

$$\tilde{g}(s) = g(-s), \quad \tilde{V}(s, z) = -V(-s, z).$$

These functions preserve (3.1), and the spatial derivative at the reflected point is $-d$. Hence $d \geq -A + c$. Therefore, if $g(t) > y$ and there exist times $t_- < t < t_+$ such that $g(t_-) < y$ and $g(t_+) < y$, then

$$-A + c \leq V_y(t, y) \leq A - c. \quad (3.5)$$

Applying spatial reflection $\hat{g} = -g$, $\hat{V}(s, z) = V(s, -z)$ gives the same pair of bounds if $g(t) < y$ and there exist times $t_- < t < t_+$ such that $g(t_-) > y$ and $g(t_+) > y$. Crucially, c depends only on K , not on the rectangle.

Step 3: finite iteration. Start with a rectangle $I_0 \times Q_0$ around $(t_*, g(t_*))$ and a bound $A_0 = M$. Suppose a bound $A_n \geq 0$ is valid on $I_n \times Q_n$. By Step 1, choose two times before t_* and two after it, all in I_n , providing both a value below and a value above $g(t_*)$ on each side. Choose an interval $Q_{n+1} \subseteq Q_n$ around $g(t_*)$ above both lower values and below both upper values. Choose $I_{n+1} \subseteq I_n$ around t_* so that the two chosen times before t_* lie to the left of I_{n+1} and the two chosen times after t_* lie to its right.

For every fixed $t \in I_{n+1}$ and almost every $y \in Q_{n+1}$, the vertical derivative exists $V_y(t, y)$ and $y \neq g(t)$. The above argument and (3.5) imply

$$-A_n + c \leq V_y(t, y) \leq A_n - c.$$

If $A_n < c$, these inequalities are inconsistent. Otherwise, absolute continuity gives the improved Lipschitz bound $A_{n+1} = A_n - c$ on $I_{n+1} \times Q_{n+1}$. After finitely many iterations, $A_n = M - nc < c$, a contradiction. $\square$

4. THE CASE $p = \infty$

We retain the notation of Section 2, now with $p = \infty$, $\alpha = 1/2$, and $|a|, |b| \leq M$.

Lemma 4.1. *The distribution $D_x u$ is a finite signed Radon measure, and*

$$D_x u = a dx dy + m(x) dx \delta_{f(x)}(dy) \quad \text{for some } m \in L^\infty(\mathbb{R}). \quad (4.1)$$

The second term is supported on Γ and has compact support.

Proof. If a horizontal segment of length $h > 0$ is disjoint from Γ , the increment of u along it is at most Mh . This follows for almost every horizontal line from the Sobolev slicing theorem, and for every such segment by continuity and approximation. If the segment meets Γ , its height belongs to the interval $f([x, x+h])$, of length at most $K\sqrt{h}$. A detour just above the maximum of f on $[x, x+h]$, together with vertical Lipschitz continuity, gives

$$|u(x+h, y) - u(x, y)| \leq Mh + 2MK\sqrt{h}. \quad (4.2)$$

Let Q be a bounded interval containing the vertical projection of $\text{supp } u$. The function $q_h u(x, \cdot)$ vanishes outside Q . For every x , integrate the bound $|q_h u| \leq M$ over those heights y for which the horizontal segment from (x, y) to $(x + h, y)$ is disjoint from Γ , and use (4.2) divided by h for $y \in f([x, x + h])$, which are precisely the heights for which this horizontal segment meets Γ . Since the length of $f([x, x + h])$ is at most $K\sqrt{h}$, this gives

$$\int_{\mathbb{R}} |q_h u(x, y)| dy \leq M \cdot \text{length}(Q) + 2MK^2 =: C_0. \quad (4.3)$$

The functions $q_h u$ also have uniformly compact horizontal support. Together with (4.3), this shows that the signed measures $q_h u dx dy$ have uniformly bounded total variation. By weak-* compactness, every sequence $h_j \downarrow 0$ has a subsequence along which $q_{h_j} u dx dy$ converges weak-* to a finite signed Radon measure. Since $q_h u \rightarrow D_x u$ in the sense of distributions, this limit is necessarily $D_x u$. Thus $D_x u$ is a finite signed Radon measure. For nonnegative $\zeta \in C_c(\mathbb{R})$, the weighted version of (4.3) and lower semicontinuity of total variation give

$$\int_{\mathbb{R}^2} \zeta(x) d|D_x u|(x, y) \leq C_0 \int_{\mathbb{R}} \zeta(x) dx.$$

Equivalently, for $\pi_x(x, y) = x$,

$$(\pi_x)_\# |D_x u| \leq C_0 dx. \quad (4.4)$$

Off the graph, $D_x u = a dx dy$. Since Γ has area zero, $\mu := D_x u - a dx dy$ is the restriction of $D_x u$ to Γ . The restriction of π_x to Γ is one-to-one, with inverse $x \mapsto (x, f(x))$. It follows from (4.4) that $(\pi_x)_\# \mu = m(x) dx$ with $m \in L^\infty$. Pulling this measure back to the graph gives (4.1). In particular, this is a measure parametrized by dx , not by arclength. $\square$

Proof of Theorem 1.1 for $p = \infty$. It suffices to prove $m = 0$ in (4.1). Otherwise, choose a Lebesgue point x_0 of m at which $m_0 := m(x_0) \neq 0$, and write $f_0 = f(x_0)$. For $r > 0$, define

$$g_r(s) = \frac{f(x_0 + r^2 s) - f_0}{r}, \quad U_r(s, z) = \frac{u(x_0 + r^2 s, f_0 + rz) - u(x_0, f_0)}{r}.$$

The Hölder bound for f , the vertical M -Lipschitz continuity for u , and (4.2) yield

$$|g_r(s) - g_r(t)| \leq K\sqrt{|s - t|},$$

$$|U_r(s, z) - U_r(t, w)| \leq M|z - w| + Mr|s - t| + 2MK\sqrt{|s - t|}.$$

Moreover, $g_r(0) = 0$ and $U_r(0, 0) = 0$. Arzelà–Ascoli and a diagonal subsequence therefore give locally uniform limits $g_r \rightarrow g$ and $U_r \rightarrow U$ on $\mathbb{R}$ and $\mathbb{R}^2$. In particular,

$$|g(s) - g(t)| \leq K\sqrt{|s - t|}, \quad \text{Lip}(U(s, \cdot)) \leq M. \quad (4.5)$$

A change of variables in (4.1) gives

$$D_s U_r = r a(x_0 + r^2 s, f_0 + rz) ds dz + m(x_0 + r^2 s) ds \delta_{g_r(s)}(dz). \quad (4.6)$$

As $r \rightarrow 0$, the first term tends to zero because $|a| \leq M$. For each fixed $R > 0$, the Lebesgue-point property gives

$$\int_{-R}^R |m(x_0 + r^2 s) - m_0| ds = \frac{1}{r^2} \int_{x_0 - Rr^2}^{x_0 + Rr^2} |m(x) - m_0| dx \rightarrow 0.$$

Using also locally uniform convergence of g_r , we may pass to the limit in (4.6) to obtain

$$D_s U = m_0 ds \delta_{g(s)}(dz).$$

Set $V = U/m_0$. Then

$$D_s V = ds \delta_{g(s)}(dz), \quad \text{Lip}(V(s, \cdot)) \leq M/|m_0|. \quad (4.7)$$

For $\varphi \in C_c^\infty(\mathbb{R})$, the continuous function $s \mapsto \int V(s, z)\varphi(z) dz$ has distributional derivative $\varphi(g(s))$. The fundamental theorem of calculus therefore gives, for every $s < t$,

$$\int_{\mathbb{R}} [V(t, z) - V(s, z)]\varphi(z) dz = \int_s^t \varphi(g(r)) dr.$$

Thus $V(t, \cdot) - V(s, \cdot)$ is a continuous density of the occupation measure on $[s, t]$. Positivity follows from positivity of that measure and continuity of the density. Equations (4.5) and (4.7) contradict Lemma 3.1 on any nondegenerate bounded time interval. Therefore $m = 0$. Now (2.1) and (4.1) imply $u \in W^{1,\infty}(\mathbb{R}^2)$, as required. $\square$

Proof of Corollary 1.2. For completeness, let F be a homeomorphism of the sphere conformal off a $2/3$ -Hölder graph Γ . Postcompose with a Möbius map so that $F(\infty) = \infty$. On a bounded neighborhood D of Γ with compact closure, injectivity and the conformal change-of-variables formula give

$$\int_{D \setminus \Gamma} |F'(z)|^2 dA(z) = |F(D \setminus \Gamma)| < \infty.$$

The real and imaginary parts of F are continuous and belong to $W^{1,2}(D \setminus \Gamma)$. Theorem 1.1 extends them across Γ . Since the graph has area zero, $\bar{\partial}F = 0$ distributionally on D . Weyl's lemma implies that F is holomorphic on D , and hence conformal on the whole sphere. Thus F is Möbius. $\square$

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